Abhilash Reddy Yammanuru

+1-4807915951 • abhilashreddyy123@gmail.com • https://www.linkedin.com/in/yabhilashreddy/
• https://github.com/abhilashreddyy • San Jose CA

Backend & ML Engineer (5+ yrs) building distributed, low-latency FastAPI service and data platforms on AWS/Kubernetes in Python and Java Script, now focused on production LLM systems with structured outputs.

PROFESSIONAL EXPERIENCE

AXYO AI, (Backend & ML Engineer) *July 2023 – July 2026*
Tempe - AZ

- Built and deployed distributed, containerized **FastAPI Restful** services on **AWS EKS (Kubernetes)** via **CI/CD**, horizontally scaled behind a load balancer for high-availability, fault-tolerant production traffic.
- Designed end-to-end data platforms in **Python** for demand/supply planning, supplier performance, and financial planning transformations over **SAP B1** and ERP sources, with **PostgreSQL** schema and KPI modeling, and **FastAPI**-served dashboards using **REST API** on **AWS**.
- Built an **LLM Powered** synthetic data generation pipeline where LLMs infer schema semantics and relationships (temporal constraints, foreign keys), with structured output validation and decision-tree-based segmentation preserving statistical properties of source data.
- Cut API latency **~3s to ~300ms (10x)** via profiling, bulk authorization checks, a multi-component fetch API, and **PostgreSQL** query optimization.
- Built and maintained **Airflow (MWAA)** pipelines with **event-driven S3-to-Lambda triggers** to decouple and coordinate work across isolated Airflow environments, monitored via **CloudWatch**.
- Enforced schema-driven **Pydantic** data contracts across APIs, pipelines, and **LLM** workflows.

Grainite Inc (Software Engineer Intern) *May 2022 – Aug 2022*
Santa Clara - CA

- Built a **Python** client SDK (**gRPC** + protocol buffers) for a **distributed streaming** / event-processing platform, enabling low-latency, high-throughput data ingestion for downstream consumers.
- Automated **API** library evaluation, testing, and packaging by integrating code into **CI/CD pipelines** using Bazel and shell scripting.
- Engineered concurrent event-loop batch processing (**multi-threading**), improving ingestion throughput.

Simbo.ai (Machine Learning Engineer) *June 2019 – June 2021*
Bengaluru - India

- Built a full-stack data annotation platform using Django, Python, and JavaScript, enabling generation of **1M+ labeled data points.**
- Designed ETL pipelines to process and store **18M+ records across 5 sources**, ensuring high-quality data for ML systems.
- Built and Deployed ML models on **AWS (EC2, S3, CloudWatch)** for scalable and reliable inference.
- Developed **NLP models** with **LSTM**-based architectures for medical text segmentation and prescription generation.

TECHNICAL SKILLS

Languages & Data: Python, SQL, C++, Java Script, Bash, PostgreSQL (+ Java/Go/Spark)
Backend, Cloud & ML: FastAPI, Django, gRPC, distributed systems, event-driven/streaming; AWS (EKS, S3, Lambda, MWAA), Kubernetes, Docker, CI/CD, Airflow, ETL; PyTorch, LLMs, NLP

EDUCATION

Masters in Computer Science *Aug 2021 – May 2023*
Arizona State University **4/4** GPA

B.Tech. Computer Science and Engineering *Jul 2015 – Jun 2019*
Savitribai Phule Pune University (SPPU) **8.52/10** GPA

PROJECTS

Jobbive - Job Matching Platform · live (Founder)

- Scrapes 300K+ live postings everyday from 2,900+ companies across 20 ATS platforms; an encoder parses each into structured requirements once, then a deterministic rubric scores them against your résumé.

LLM Inference Benchmarking & Optimization · github · live dashboard

- Benchmarks LLMs on **HumanEval** and **GSM8K** across inference optimizations (***quantization, KV-cache, batching, speculative decoding***); live dashboard tracks throughput, latency, and memory.